PINAKI IT CONSULTANT PRIVATE LIMITED

FREE SAMPLE NOTES

MICROSOFT EXCEL BASICS

A Beginner's Guide to Spreadsheets, Formulas, and Data Analysis

B5	*fx*	=SUM(B2:B4)		
	A	B	C	D
1	**Month**	**Sales**	**Target**	**Variance**
2	Jan	12,000	15,000	-3,000
3	Feb	15,000	15,000	0
4	Mar	18,000	15,000	3,000
5	Total	**45,000**	45,000	0

ABOUT THESE NOTES

Microsoft Excel is the most widely used tool for organising, calculating and analysing data, and it is the natural first step into the world of data analytics. These notes take you from your very first workbook to a complete mini project, using simple examples with students, employees and monthly sales.

KEY TOPICS COVERED

- What Excel is and how the interface works
- Workbooks, data entry and formatting
- Formulas and cell references
- SUM, AVERAGE, MIN, MAX, COUNT and COUNTA
- IF, AND, OR, COUNTIF and SUMIF
- Sorting, filtering and finding data
- Data cleaning and data validation
- Conditional formatting and Excel Tables
- Charts and data visualisation
- A complete Student Performance mini project

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1. Introduction to Microsoft Excel

1.1 What Is Microsoft Excel?

Microsoft Excel is a **spreadsheet application** used to organise, calculate, analyse and visualise data. Information is stored in a grid of rows and columns, and **formulas** recalculate automatically whenever the data changes. This means you build a worksheet once and reuse it again and again.

1.2 Why Is Excel Important?

Area	Typical use in Excel	Area	Typical use in Excel
Data analytics	Cleaning and exploring data sets	**Inventory**	Stock levels and reorder alerts
Accounting & finance	Budgets, invoices, expense tracking	**Education**	Marks, attendance and grade sheets
Human resources	Employee records, attendance, payroll	**Business reporting**	Summaries and charts for managers
Sales reporting	Monthly sales, targets, commission		

1.3 Key Features of Excel

- **Data entry** for text, numbers and dates
- **Formulas and functions** for calculations
- **Sorting and filtering** to organise records
- **Tables** for structured, expandable data
- **Charts** to turn numbers into pictures
- **Conditional formatting** to highlight values
- **PivotTables** (preview): summarise large data sets quickly. Covered in the complete notes.

1.4 The Excel Interface

The picture below shows the main parts of an Excel window. The selected cell is **B3**: column **B**, row **3**.

B3 | *fx* | 65

	A	B	C	D
1	**Student**	**Marks**	**Grade**	
2	Aman	78	B	
3	Bhavna	**65**	C	
4	Chetan	90	A	

Sheet1 | Sheet2 | + Add new sheet (sheet tabs)

Term	Meaning	Term	Meaning
Workbook	An Excel file (.xlsx) that contains one or more worksheets	**Cell address**	Column letter + row number, such as B3
Worksheet	A single page of cells inside a workbook	**Formula bar**	Shows and edits the contents of the selected cell
Row	Horizontal line of cells, numbered 1, 2, 3, …	**Ribbon**	Tabs (Home, Insert, Data …) holding the commands
Column	Vertical line of cells, lettered A, B, C, …	**Name Box**	Shows the address of the selected cell; type an address to jump there
Cell	One box where a row and column meet; holds data	**Sheet tabs**	Tabs at the bottom to switch between worksheets

Remember: a cell address always gives the **column first, then the row**. So B3 means column B, row 3, and in the example above it holds the value 65. Instructions in these notes apply to desktop Excel 2016, 2019, 2021 and Microsoft 365.

2. Workbooks, Data Entry, and Formatting

A **workbook** is the whole Excel file, like a notebook. A **worksheet** is one page (tab) inside it. One workbook can hold many worksheets, for example one worksheet for each month of the year.

2.1 Working with Workbooks and Worksheets

Task	How to do it
Create a new workbook	**File > New > Blank workbook** (Ctrl + N)
Open and save	**File > Open** (Ctrl + O). To save, press Ctrl + S, choose a folder, type a file name and keep the **.xlsx** format.
Rename a worksheet	Double-click the sheet tab, type the new name and press Enter.
Add or delete a sheet	Click the **+** button next to the sheet tabs to add one. To delete, right-click the tab and choose **Delete**. This cannot be undone, so save a copy first.

2.2 Entering, Editing and Selecting Data

- **Enter data:** click a cell, type, then press Enter. Text aligns left by default; numbers and dates align right.
- **Edit a cell:** double-click it or press F2. Press Enter to confirm or Esc to cancel.
- **Select cells:** click one cell, or drag to select a range such as `A1:C5`. Hold Ctrl to select separate cells. Click a column letter or row number to select the whole column or row.
- **Copy and paste:** Ctrl + C then Ctrl + V. Use Ctrl + X to move (cut) data instead.
- **Row height and column width:** drag the border between headings, or double-click the border to AutoFit to the content.

2.3 Basic Cell Formatting

Most formatting tools are on the **Home** tab, in the Font, Alignment and Number groups.

Option	What it does
Font style and size	Changes how text looks. Use a clear font, such as Calibri 11. Bold, italic and underline (Ctrl + B, I, U) emphasise headings.
Fill and font colour	Shades the cell background or changes the text colour. Use sparingly.
Borders and alignment	Borders draw lines around cells. Alignment sets left, centre or right position.
Number format	Shows the same value as a number, **currency** (for example 1,500.00), **percentage** (0.78 becomes 78%) or date.

Example: Before and After Formatting

BEFORE (unformatted)

student	marks	percent
ravi kumar	78	0.78
sneha rao	65	0.65
amit shah	90	0.9

AFTER (formatted)

Student	Marks	Percent
Ravi Kumar	78	78%
Sneha Rao	65	65%
Amit Shah	90	90%

Useful Keyboard Shortcuts

Shortcut	Action	Shortcut	Action
`Ctrl + C`	Copy	`Ctrl + S`	Save the workbook
`Ctrl + V`	Paste	`Ctrl + B`	Bold
`Ctrl + Z`	Undo the last action	`Ctrl + F`	Find

Shortcuts can vary slightly between operating systems and Excel versions. On a Mac, use the Command key in place of Ctrl for most of them.

3. Understanding Formulas and Cell References

Theory. A **formula** is an instruction that tells Excel to calculate something. Every formula starts with an **equals sign (=)**. The = sign is how Excel knows the cell contains a calculation and not plain text. Without it, `B2+C2` is just text.

3.1 Your First Formulas

Here is a small marks sheet. Column D holds the total and column E holds the average.

	A	B	C	D	E
1	**Student**	**Subject 1**	**Subject 2**	**Total**	**Average**
2	Student A	78	85	163	81.5
3	Student B	65	72	137	68.5
4	Student C	90	88	178	89

Cell	Formula	Result	Explanation
D2	`=B2+C2`	163	Adds Subject 1 (78) and Subject 2 (85).
E2	`=(B2+C2)/2`	81.5	Adds first, then divides by 2. The brackets make Excel add before dividing.

Operators: `+` add, `-` subtract, `*` multiply, `/` divide, `^` power. For example `=10+5` gives 15, `=10*5` gives 50 and `=2^3` gives 8.

> **Why brackets matter:** `=B2+C2/2` gives 120.5 because Excel divides first (85 / 2 = 42.5, then 78 + 42.5). `=(B2+C2)/2` gives the correct average, 81.5.

3.2 Cell References, Relative, Absolute and Mixed

You could type `=78+85`, but that locks the numbers inside the formula. If you write `=B2+C2` instead, Excel reads whatever is in B2 and C2. Change B2 to 80 and the total updates to **165** automatically. Always prefer cell references.

When you copy a formula to another cell, Excel adjusts the references. Copy `=B2+C2` from D2 down to D3 and it becomes `=B3+C3`. This is a **relative reference**. Sometimes you want a reference to stay fixed. Put a **$** sign in front of the column letter, the row number, or both.

Type	Example	What happens when the formula is copied
Relative	A1	Column and row both change (A1 becomes A2, B1 ...).
Absolute	A1	Never changes. It always points to cell A1.
Mixed (column fixed)	$A1	Column A stays fixed; the row number changes.
Mixed (row fixed)	A$1	Row 1 stays fixed; the column letter changes.

Practical use. Tax is stored once in cell **F1** (18%). The absolute reference `$F$1` makes every copied formula use that same cell. Press **F4** after typing a reference to add the $ signs quickly.

	A	B	C	D	E	F
1	**Item**	**Price**	**Tax**		**Tax rate**	**18%**
2	Notebook	200	36			
3	Pen pack	150	27			
4	Folder	450	81			

FORMULA IN C2, THEN COPY DOWN TO C4 (RESULTS 36, 27 AND 81)

`=B2*$F$1`

> **Without the $ signs:** copying `=B2*F1` to C3 gives `=B3*F2`. Cell F2 is empty, so the tax becomes 0. This is the most common beginner mistake, and it is fixed by using `$F$1`.

4. Basic Mathematical and Statistical Functions

Theory. A **function** is a ready-made formula with a name. Instead of writing `=B2+B3+B4+B5+B6`, you write `=SUM(B2:B6)`. The part inside the brackets is the **range**: `B2:B6` means "from B2 through B6".

4.1 Syntax

GENERAL PATTERN

```
=SUM(number1, [number2], ...)
=COUNTA(value1, [value2], ...)
```

AVERAGE, MIN, MAX and COUNT follow the same pattern as SUM. Square brackets mean the argument is optional. Examples use commas as separators; some regional settings use semicolons (`;`) instead.

4.2 Example Data and Reference Table

	A	B
1	**Student**	**Marks**
2	Aman	78
3	Bhavna	65
4	Chetan	90
5	Divya	55
6	Esha	82

All examples below use the marks stored in **B2:B6** and the names in **A2:A6**. These five marks add up to 370.

> **Tip:** The status bar at the bottom of Excel shows Sum, Average and Count instantly when you select cells.

Function	Purpose	Example	Result	How it works
SUM	Adds numbers	`=SUM(B2:B6)`	**370**	78 + 65 + 90 + 55 + 82 = 370.
AVERAGE	Finds the mean	`=AVERAGE(B2:B6)`	**74**	Total divided by how many numbers: 370 / 5 = 74.
MIN	Smallest value	`=MIN(B2:B6)`	**55**	Divya has the lowest mark.
MAX	Largest value	`=MAX(B2:B6)`	**90**	Chetan has the highest mark.
COUNT	Counts cells containing numbers	`=COUNT(B2:B6)`	**5**	Five cells contain numbers.
COUNTA	Counts cells that are not empty	`=COUNTA(A2:A6)`	**5**	Five names are present. COUNTA counts text too.

4.3 COUNT versus COUNTA

COUNT counts **numbers only**. COUNTA counts **every non-empty cell**, whether it holds numbers, text or dates. Suppose Chetan was absent and "Absent" was typed in B4:

	A	B
1	**Student**	**Marks**
2	Aman	78
3	Bhavna	65
4	Chetan	Absent
5	Divya	55
6	Esha	82

Formula	Result	Why
`=COUNT(B2:B6)`	**4**	"Absent" is text, so it is skipped.
`=COUNTA(B2:B6)`	**5**	All five cells contain something.

Use COUNT to see how many marks were entered, and COUNTA to see how many records exist.

> **Real-world use:** SUM totals monthly expenses, AVERAGE finds average sales per day, MAX and MIN spot the best and worst performer, and COUNTA counts how many employees are listed.

5. Logical, Conditional and Text Functions

5.1 IF, AND and OR

Theory. **IF** tests a condition and returns one result when it is true and another when it is false. **AND** is true only when all conditions are true. **OR** is true when at least one condition is true.

SYNTAX

```
=IF(logical_test, value_if_true, value_if_false)
=AND(condition1, condition2, ...)      =OR(condition1, condition2, ...)
```

	A	B	C	D	E	F
1	**Student**	**Maths**	**Science**	**IF**	**AND**	**OR**
2	Aman	78	85	Pass	Pass	Yes
3	Bhavna	35	72	Fail	Fail	No
4	Chetan	90	38	Pass	Fail	Yes
5	Divya	55	61	Pass	Pass	No

Cell	Formula (row 2)	Meaning
D2	`=IF(B2>=40,"Pass","Fail")`	If Maths is 40 or more, show Pass; otherwise show Fail.
E2	`=IF(AND(B2>=40,C2>=40),"Pass","Fail")`	Pass only when **both** Maths and Science are at least 40. Chetan fails because his Science mark is 38.
F2	`=IF(OR(B2>=85,C2>=85),"Yes","No")`	Yes if **either** subject is 85 or more (a simple scholarship check).

5.2 COUNTIF and SUMIF

Theory. **COUNTIF** counts the cells that meet a condition; **SUMIF** adds the values that meet a condition. Text and operators in a condition go inside quotation marks, such as `">=40"`.

Syntax. `=COUNTIF(range, criteria)` and `=SUMIF(range, criteria, [sum_range])`

	A	B
1	**Student**	**Maths**
2	Aman	78
3	Bhavna	35
4	Chetan	90
5	Divya	55

Formula	Result
`=COUNTIF(B2:B5,">=40")`	**3**

Three students scored 40 or more: 78, 90 and 55.

	A	B
1	**Dept**	**Salary**
2	Sales	30,000
3	HR	28,000
4	Sales	35,000
5	IT	40,000
6	Sales	32,000

Formula	Result
`=SUMIF(A2:A6,"Sales",B2:B6)`	**97,000**

Adds only the Sales salaries: 30,000 + 35,000 + 32,000.

5.3 CONCAT: Combining Text

Theory. CONCAT joins text from several cells into one. With "Aman" in A2 and "Sharma" in B2:

Formula	Result	Explanation
`=CONCAT(A2," ",B2)`	**Aman Sharma**	The " " in the middle adds a space between the names.

CONCAT needs Excel 2019 or Microsoft 365; older versions use `CONCATENATE`, which works the same way. The `&` operator, as in `=A2&" "&B2`, gives the same result in every version.

6. Sorting, Filtering and Finding Data

Large data sets are hard to read. **Sorting** rearranges rows into an order. **Filtering** shows only the rows that match a condition. Both are on the **Data** tab.

	Sorting	Filtering
What it does	Changes the order of the rows	Shows only matching rows and hides the rest
Data affected?	All rows stay; only their order changes	Hidden rows are **not deleted**; clear the filter to see them again
Example	Marks from highest to lowest	Only students with marks above 75

6.1 Sorting

- Click any cell in the column you want to sort by (do not select just one column).
- On the **Data** tab choose **A to Z** or **Z to A** for text, and **Smallest to Largest** or **Largest to Smallest** for numbers.
- Excel moves the whole row, so each student keeps their own marks. If asked, choose **Expand the selection**.

Before sorting

	A	B
1	**Name**	**Marks**
2	Aman	82
3	Bhavna	68
4	Chetan	91
5	Divya	74

After sorting Marks: Largest to Smallest

	A	B
1	**Name**	**Marks**
2	Chetan	91
3	Aman	82
4	Divya	74
5	Bhavna	68

6.2 Filtering: Students Who Scored More Than 75

1. Click any cell inside the data.
2. Select **Data > Filter** (or Ctrl + Shift + L). Drop-down arrows appear in the headings.
3. Click the arrow in the **Marks** heading and choose **Number Filters > Greater Than...**
4. Type **75** and click **OK**. To remove the filter, choose **Data > Clear**.

Original data

	A	B	C
1	**ID**	**Name**	**Marks**
2	S01	Aman	82
3	S02	Bhavna	68
4	S03	Chetan	91
5	S04	Divya	74
6	S05	Esha	77
7	S06	Farhan	59

After filtering Marks > 75

	A	B	C
1	**ID**	**Name**	**Marks**
2	S01	Aman	82
4	S03	Chetan	91
6	S05	Esha	77

Rows 3, 5 and 7 are hidden, not deleted. Notice the row numbers skip.

6.3 Finding Data and Removing Duplicates

- **Find:** press **Ctrl + F**, type the value and press Enter. Use **Find All** to see every match.
- **Remove duplicates:** select the data, choose **Data > Remove Duplicates**, tick the columns to compare and click OK. Unlike a filter, this **permanently deletes** repeated rows, so work on a copy.

> **Remember:** filtering hides rows that do not match; it does not delete them. The data returns as soon as you clear the filter. Sorting only changes the order of the rows.

7. Data Cleaning and Basic Data Validation

Theory. Clean data gives correct answers. A department typed as `Sales ` (with a trailing space) will not match `Sales` in a filter or COUNTIF, and a name typed as "rahul SHARMA" looks unprofessional in a report. Always check your data before analysing it.

7.1 Common Problems and Fixes

Problem	How to find or fix it
Blank cells	**Home > Find & Select > Go To Special > Blanks** selects every empty cell so you can fill or review them.
Duplicate records	Highlight them with **Conditional Formatting > Highlight Cells Rules > Duplicate Values**, or remove them with **Data > Remove Duplicates**.
Extra spaces	Use the `TRIM` function (below).
Inconsistent text	Use `UPPER`, `LOWER` or `PROPER`. Use **Find and Replace** (Ctrl + H) to correct entries such as "Bombay" to "Mumbai".
Inconsistent dates	Select the dates, press **Ctrl + 1** and choose one **Date** format. True dates align to the right.

7.2 Text Functions for Cleaning

SYNTAX

```
=TRIM(text)   =UPPER(text)   =LOWER(text)   =PROPER(text)
```

Cell contents	Formula	Result	What it does
" Amit Verma "	`=TRIM(A2)`	**Amit Verma**	Removes extra spaces at the start, end and between words.
"priya SINGH"	`=UPPER(A3)`	**PRIYA SINGH**	Converts all letters to capitals.
"priya SINGH"	`=LOWER(A3)`	**priya singh**	Converts all letters to small letters.
"priya SINGH"	`=PROPER(A3)`	**Priya Singh**	Capitalises the first letter of each word.

Practical example. Functions can be combined. `=PROPER(TRIM(A2))` first removes the spaces, then fixes the capitals:

Raw name (column A)	Cleaned with =PROPER(TRIM(A2))	Copied down
" rahul SHARMA "	**Rahul Sharma**	rows 2, 3 and 4
"NEHA gupta"	**Neha Gupta**	
"amit verma"	**Amit Verma**	

To keep the cleaned names, copy them and use **Paste Special > Values** over the original column. This replaces the formulas with plain text.

7.3 Data Validation: A Drop-Down List

Theory. **Data Validation** limits what can be typed into a cell. A drop-down list lets users pick from approved choices, so nobody can mistype a department. This keeps filters, counts and totals reliable.

1. Select the cells to control, for example C2:C20.
2. Choose **Data > Data Validation**. On the **Settings** tab set **Allow** to **List**.
3. In **Source** type `HR,Sales,IT,Finance` (use semicolons if your regional settings require them) and click **OK**.

Looking ahead: **Power Query** is an advanced Excel feature for cleaning very large or messy data automatically. You will meet it in the complete notes.

8. Conditional Formatting and Excel Tables

8.1 Conditional Formatting

Theory. **Conditional formatting** changes the look of a cell automatically, based on its value. It lets you spot high, low or repeated values at a glance, and it updates itself when the data changes.

Rule type	What it does
Highlight Cells Rules	Colours cells that are greater than, less than, between or equal to a value, or that are duplicates.
Color Scales	Shades cells from one colour to another (for example light to dark) so you can see high and low values.
Data Bars	Draws a bar inside each cell whose length matches the value.

Example: Highlight Marks Below 40 and Above 80

1. Select the marks, for example **B2:B6**.
2. Choose **Home > Conditional Formatting > Highlight Cells Rules > Less Than...**
3. Type **40**, pick **Light Red Fill with Dark Red Text** and click OK.
4. Repeat with **Greater Than...**, value **80**, and choose **Green Fill with Dark Green Text**.

To remove rules: **Conditional Formatting > Clear Rules**.

	A	B
1	**Student**	**Marks**
2	Aman	82
3	Bhavna	35
4	Chetan	91
5	Divya	74
6	Esha	38

Green = above 80, red = below 40.

> **Practical use:** a teacher can see instantly that Bhavna and Esha need extra support, and a sales manager can highlight every record that falls below the monthly target.

8.2 Excel Tables

Theory. An **Excel Table** turns a plain range into a structured object with its own style, built-in filter buttons and automatic expansion.

1. Click any cell inside your data (the first row should contain headings).
2. Choose **Insert > Table** (or press **Ctrl + T**). Confirm the range and tick **My table has headers**, then click OK.
3. Use the **Table Design** tab to pick a **Table Style** and give the table a name.
4. Type a new record directly below the last row. The table **expands automatically** and copies the formatting and formulas.

- **Benefits:** banded rows for readability, filter arrows already in place, automatic growth, and a Total Row option on the Table Design tab.

	Table formatting	Conditional formatting
Purpose	Gives the whole data set a clean, structured look	Highlights particular values that meet a rule
Changes with data?	No. The style stays the same	Yes. Colours update when values change

This sample does not cover structured-reference formulas for tables. They are part of the complete notes.

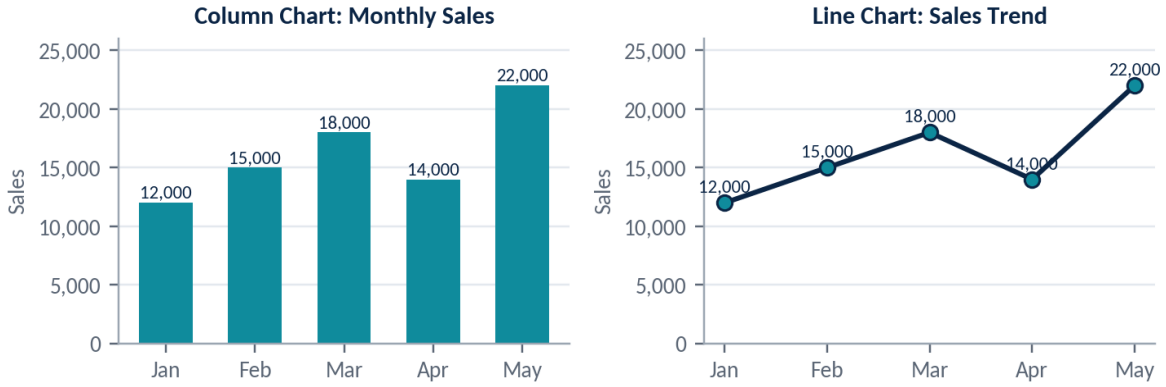
9. Charts and Data Visualization

Theory. A chart turns rows of numbers into a picture, so trends and differences become obvious. Managers read a chart in seconds; a table of numbers can take minutes.

9.1 Create a Chart in Six Steps

	A	B
1	**Month**	**Sales**
2	January	12,000
3	February	15,000
4	March	18,000
5	April	14,000
6	May	22,000

1. **Select the data**, including the headings (A1:B6).
2. Open the **Insert** tab and go to the **Charts** group.
3. **Choose a chart type**, for example Clustered Column or Line.
4. Click the chart title and type a clear title, such as "Monthly Sales".
5. Add **data labels** when exact values help (Chart Design > Add Chart Element, or the + button).
6. **Format the chart:** change the style or colours and label the axes.



Both charts use the same data. The line makes the dip in April and the peak in May easy to see.

9.2 Choosing the Right Chart

Chart	Best used for	Example
Column	Comparing values across categories	Sales for each month
Bar	Comparing categories, especially with long names	Marks of students, sales by product
Line	Showing change or **trends over time**	Sales from January to May
Pie	Showing parts of a whole (a few slices only)	Share of sales by region

Best choice for sales trends: a **line chart** (or column chart). Use pie charts carefully: they work only for a small number of parts that add up to a whole, and they become hard to read with many slices.

10. Useful Functions and Productivity Shortcuts

These everyday tools save time in real spreadsheet work. The table uses an employee code **EMP1024** in cell A2, a value of 3.14159 in B2, and the dates 01-Oct-2026 (C2) and 15-Oct-2026 (D2).

Function	Purpose	Example	Result
TODAY()	Returns the current date; it updates every day. No arguments needed.	`=TODAY()`	Today's date
ROUND()	Rounds a number to a set number of decimal places.	`=ROUND(B2,2)`	**3.14**
LEFT()	Takes characters from the start of a text value.	`=LEFT(A2,3)`	**EMP**
RIGHT()	Takes characters from the end of a text value.	`=RIGHT(A2,4)`	**1024**
LEN()	Counts the characters in a text value, including spaces.	`=LEN(A2)`	**7**
Date difference	Subtract one date from another to get the number of days.	`=D2-C2`	**14**

SYNTAX

```
=ROUND(number, num_digits)
=LEFT(text, [num_chars])    =RIGHT(text, [num_chars])    =LEN(text)
```

If `=D2-C2` shows a date instead of 14, format the result cell as **General** or **Number** on the Home tab. ROUND changes the stored value, while number formatting only changes how it looks.

10.1 Find and Replace

Press **Ctrl + H**, type the text to find in **Find what** and the new text in **Replace with**. Click **Replace All** to fix every match, or **Replace** one at a time. Useful for correcting repeated spelling mistakes.

10.2 AutoFill

The **fill handle** is the small square at the bottom-right corner of a selected cell. Drag it to copy values or formulas, or to continue a pattern.

- Type **Jan** and drag: Excel fills Feb, Mar, Apr and so on. Type **1** and **2**, select both and drag to continue 3, 4, 5.
- **Double-click** the fill handle to copy a formula down as far as the neighbouring data goes.

10.3 More Shortcuts

Shortcut	Action	Shortcut	Action
`Ctrl + Y`	Redo	`Ctrl + T`	Create a table
`Ctrl + A`	Select all cells	`Ctrl + Shift + L`	Turn filters on or off
`Ctrl + Home`	Jump to cell A1	`Ctrl + 1`	Open Format Cells
`F2`	Edit the selected cell	`Alt + =`	AutoSum selected cells

11. Mini Project: Student Performance Analysis

Objective. Use Excel to organise student marks, calculate results, spot performance patterns and build a simple report. Every subject is scored out of 100. **Passing rule: a student passes only if the marks in all three subjects are at least 33.**

	A	B	C	D	E	F	G	H
1	Student ID	Student Name	Maths	Science	English	Total	Average	Result
2	S101	Aarav Mehta	78	85	72	235	78.33	Pass
3	S102	Bhavna Rao	65	58	70	193	64.33	Pass
4	S103	Chetan Joshi	92	88	95	275	91.67	Pass
5	S104	Divya Nair	45	30	52	127	42.33	Fail
6	S105	Esha Gupta	70	76	81	227	75.67	Pass
7	S106	Farhan Ali	33	41	38	112	37.33	Pass
8	S107	Gauri Patel	28	55	60	143	47.67	Fail
9	S108	Harsh Verma	88	79	84	251	83.67	Pass

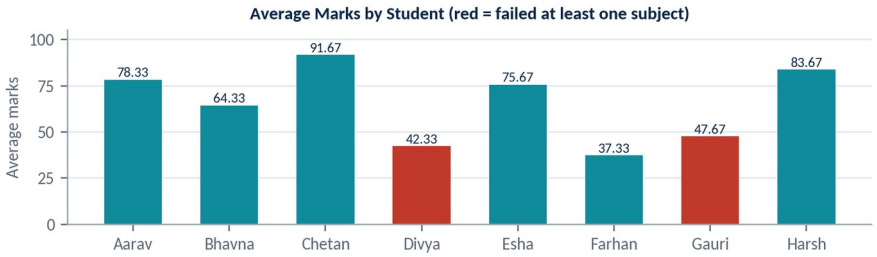
11.1 Step-by-Step Instructions (final sheet above, red = marks below 33)

#	Task	Instruction and formula
1	**Enter the data**	Type the headings in A1:H1 and the ID, name and three marks in A2:E9.
2	**Total marks**	In **F2** type `=SUM(C2:E2)`, then double-click the fill handle to copy it down to F9.
3	**Average marks**	In **G2** type `=AVERAGE(C2:E2)` and fill down; show 2 decimals.
4	**Pass or Fail**	In **H2** type `=IF(AND(C2>=33,D2>=33,E2>=33),"Pass","Fail")` and fill down. Shorter: `=IF(MIN(C2:E2)>=33,"Pass","Fail")`.
5	**Sort by average**	Click a cell in column G, then **Data > Sort Largest to Smallest**.
6	**Filter passed students**	**Data > Filter**, open the Result drop-down, tick **Pass** only and click OK.
7	**Highlight low marks**	Select **C2:E9**, then **Home > Conditional Formatting > Highlight Cells Rules > Less Than**, enter **33**.
8	**Create a chart**	Select **B1:B9**, hold Ctrl and select **G1:G9**, then **Insert > Clustered Column**. Add a chart title.

Expected results for representative students

Student	Marks (M, S, E)	Total	Average	Result	Why
Aarav Mehta	78, 85, 72	235	78.33	Pass	Every subject is above 33.
Divya Nair	45, 30, 52	127	42.33	Fail	Science is 30, which is below 33.
Farhan Ali	33, 41, 38	112	37.33	Pass	Maths is exactly 33, which still passes (>= 33).

After sorting by average (highest first): Chetan, Harsh, Aarav, Esha, Bhavna, Gauri, Divya, Farhan. **After filtering** for Pass, 6 students remain; Divya and Gauri are hidden.



What you learn: organising data in a clean table; calculating with `SUM`, `AVERAGE` and `IF`; applying one rule to several columns with `AND` (or `MIN`); and sorting, filtering, highlighting and charting data to build a report.

12. Practice, Quick Revision, and What's Next

12.1 Practice Exercises

1. Create a student marks spreadsheet with at least 6 students.
2. Calculate the total and average marks for each student.
3. Find the highest and lowest marks in a column.
4. Count the number of students in your data set.
5. Use IF() to show Pass or Fail for each student.
6. Use COUNTIF() to count students scoring at least 75.
7. Sort the students by marks, from highest to lowest.
8. Add a Department column and filter the records by department.
9. Apply conditional formatting to highlight low marks.
10. Create a chart showing monthly sales for six months.

Try these on your own first. Practising is the fastest way to learn Excel.

12.2 Quick Revision

Function	Description	Function	Description
`SUM()`	Adds numbers or a range	`AND()`	True only if all conditions are true
`AVERAGE()`	Finds the mean of the numbers	`OR()`	True if any condition is true
`MIN()`	Returns the smallest value	`COUNTIF()`	Counts cells that meet a condition
`MAX()`	Returns the largest value	`SUMIF()`	Adds values that meet a condition
`COUNT()`	Counts cells with numbers	`TRIM()`	Removes extra spaces from text
`COUNTA()`	Counts all non-empty cells	`TODAY()`	Returns the current date
`IF()`	Returns one result if true, another if false	`ROUND()`	Rounds to a set number of decimals

12.3 What's Next?

Excel is the first step into **data analytics**. You clean data (TRIM), summarise it (SUMIF, PivotTables), and present it (charts). The next tools in our series build on these skills: **SQL** stores and queries large databases, **Python** analyses data at scale, and **Power BI** creates interactive business dashboards.

1. Advanced Excel
2. VLOOKUP and XLOOKUP
3. INDEX and MATCH
4. PivotTables and PivotCharts
5. Advanced formulas
6. Dashboards and reporting
7. Macros and VBA
8. SQL and Database Management
9. Python for Data Analytics
10. Power BI and Business Intelligence

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